Meiosis – one cell becomes four sex cells 1a

Meiosis is a special type of cell division that occurs in the sex organs of animals and plants; for example, in the ovaries and testes of animals. Meiosis results in the production of *ova*, which are eggs (singular: *ovum*), in females and sperm in males. Ova and sperm are also called gametes. Haploid gametes are produced from diploid cells.

Meiosis requires two cell divisions which result in four daughter cells. Each daughter cell has half the number of chromosomes as the parent cell (daughter cells are haploid, parent cell is diploid), therefore they are not identical to the parent cell nor to each other. Meiosis is often called the **reduction division**.

Meiosis – stage 1

1 The text below describes what is occurring in each part of the following diagram.
- Use the word bank to complete the descriptions.
- Then match each description to the correct part of the diagram by writing the appropriate letter in the grey box. One has been done for you.

Word bank
- spindle
- chromosome
- Nuclear
- pair
- separated
- cell
- cell
- apart
- homologous

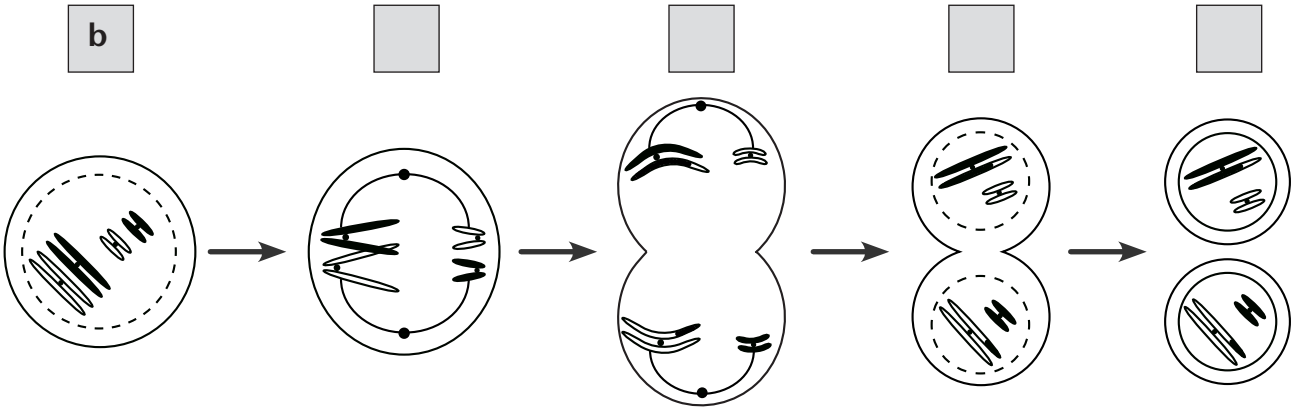
a Two new cells have formed. Each cell contains only one _ _ _ _ _ _ _ _ _ _ from each homologous pair.

b Spindle begins to appear. _ _ _ _ _ _ _ membrane begins to disappear. Chromosome doubles pair up with another set of chromosome doubles of the same size. These are called _ _ _ _ _ _ _ _ _ _ chromosomes.

c Chromosomes are now completely _ _ _ _ _ _ _ _ _. Nuclear membrane begins to form around the chromosomes. The _ _ _ _ _ _ _ begins to disappear. The cell begins to separate.

d The homologous pair of chromosomes begins to move _ _ _ _ _, to the opposite sides of the _ _ _ _.

e The homologous chromosomes line up at the centre of the _ _ _ _. Each _ _ _ _ of chromosomes is attached to the spindle.



Meiosis – stage 2

2 The text below describes what is occurring in each part of the diagram on the next page.
- Use the word bank to complete the descriptions.
- Then match each description to the correct part of the diagram by writing the appropriate letter in the grey box.

Word bank
- disappear
- split
- spindle
- separate
- division
- cell
- parent
- membrane
- chromosome

a Cell _ _ _ _ _ _ _ _ is complete. Four new cells have formed. Each cell carries half the number of chromosomes of the _ _ _ _ _ _ cell.

b Chromosomes begin to _ _ _ _ _ and move apart.

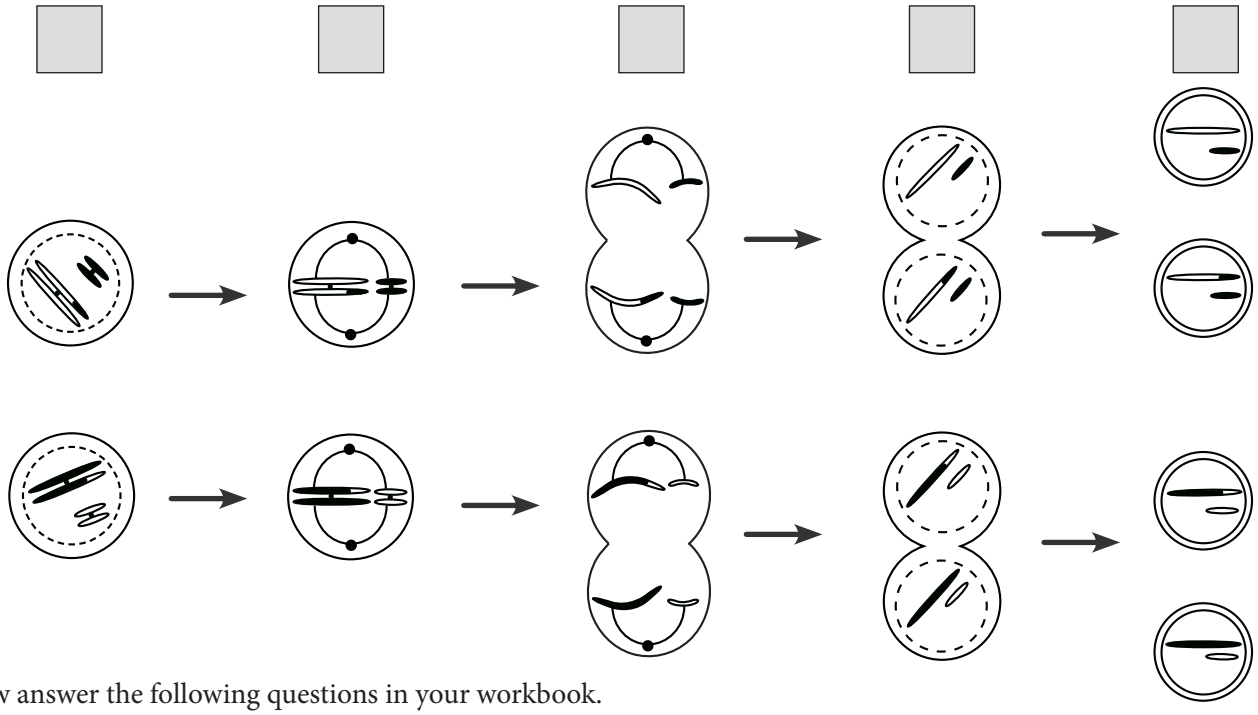
c At the beginning of stage 2, the _ _ _ _ _ _ _ reappears and the nuclear _ _ _ _ _ _ _ _ begins to disappear.

© Suada Bilali, Chris Greef and Blake Education 2010

Meiosis – one cell becomes four sex cells

d Chromosomes move towards the poles and are completely separated. The spindle begins to _ _ _ _ _ _ _ _ _. Nuclear membrane begins to form. The cell begins to _ _ _ _ _ _ _ _.

e Chromosomes line up at the equator of the _ _ _ _. Each _ _ _ _ _ _ _ _ _ _ is attached to the spindle.



Now answer the following questions in your workbook.

3 Describe the differences between stage 1 and stage 2 meiosis.

4 Define each term.

a Meiosis **b** Homologous chromosomes **c** Equator
d Poles **e** Haploid **f** Diploid

5 What do you think would happen at fertilisation if the chromosomes in the sex cells didn't halve in number during meiosis?

6 Explain why **crossing over** is so important. *Hint:* Crossing over of chromosomes occurs in the second diagram.

7 **a** In stage 2, how many sex cells were produced?
b Are they all similar?
c Why is this significant?

8 **Modelling meiosis**

- Your teacher will supply you with 2 different coloured pieces of plasticine or modelling clay.
- Construct two large pairs of chromosomes, one pair of each colour, and two small pairs of chromosomes, one pair of each colour, as is shown in the first part of the Stage 1 Meiosis diagram on the previous page.
- Use your plasticine chromosomes to reconstruct meiosis as shown in the Meiosis stage 1 and stage 2 diagrams. Move your model chromosomes to show what is occurring in each diagram, until you reach the last diagram.

a Did you end up with 4 different gametes at the end of the activity?
b What helped to increase the differences in the chromosomes?
c What was useful about using this model? For example, did it help you understand meiosis?
d What were the limitations of using this model?

© Suada Bilali, Chris Greef and Blake Education 2010